

1. Program to Count Number of Vowels Using a Separate Function

```
#include <stdio.h>
```

```
int countVowels(char str[]) {
```

```
    int i = 0, count = 0;
```

```
    while (str[i] != '\0') {
```

```
        char ch = str[i];
```

```
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
```

```
            ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {
```

```
            count++;
```

```
        }
```

```
        i++;
```

```
    }
```

```
    return count;
```

```
}
```

```
int main() {
```

```
    char str[100];
```

```
    printf("Enter a string: ");
```

```
    fgets(str, sizeof(str), stdin);
```

```
    printf("Number of vowels = %d", countVowels(str));
```

```
    return 0;
}
```

2. Function to Reverse a String Without Using strrev()

```
#include <stdio.h>
```

```
void reverseString(char str[]) {
```

```
    int i = 0, len = 0;
```

```
    char temp;
```

```
    while (str[len] != '\0') {
```

```
        len++;
```

```
    }
```

```
    len--;
```

```
    for (i = 0; i < len; i++, len--) {
```

```
        temp = str[i];
```

```
        str[i] = str[len];
```

```
        str[len] = temp;
```

```
    }
```

```
}
```

```
int main() {
```

```
    char str[100];
```

```
    printf("Enter a string: ");
```

```
    scanf("%s", str);
```

```
reverseString(str);
```

```
printf("Reversed string: %s", str);
```

```
return 0;
```

```
}
```

3. Program to Check Whether a String is Palindrome Using Functions

```
#include <stdio.h>
```

```
int isPalindrome(char str[]) {
```

```
    int i = 0, len = 0;
```

```
    while (str[len] != '\0') {
```

```
        len++;
```

```
    }
```

```
    len--;
```

```
    while (i < len) {
```

```
        if (str[i] != str[len]) {
```

```
            return 0;
```

```
        }
```

```
        i++;
```

```
        len--;
```

```
    }
```

```
    return 1;
```

```
}
```

```
int main() {
```

```
    char str[100];
```

```
    printf("Enter a string: ");
```

```
    scanf("%s", str);
```

```
    if (isPalindrome(str)) {
```

```
        printf("The string is a palindrome");
```

```
    } else {
```

```
        printf("The string is not a palindrome");
```

```
    }
```

```
    return 0;
```

```
}
```

4. Function to Calculate Length of a String Manually

```
#include <stdio.h>
```

```
int stringLength(char str[]) {
```

```
    int length = 0;
```

```
    while (str[length] != '\0') {
```

```
        length++;
```

```
    }
```

```
    return length;
```

```
}
```

```
int main() {  
    char str[100];  
  
    printf("Enter a string: ");  
    scanf("%s", str);  
  
    printf("Length of string = %d", stringLength(str));  
  
    return 0;  
}
```

5. Function to Count Number of Words in a Sentence

```
#include <stdio.h>
```

```
int countWords(char str[]) {  
    int i = 0, words = 0;  
  
    while (str[i] != '\0') {  
  
        if ((i == 0 && str[i] != ' ') ||  
            (str[i] != ' ' && str[i - 1] == ' ')) {  
            words++;  
        }  
  
        i++;  
    }  
  
    return words;
```

```
}
```

```
int main() {
```

```
    char str[200];
```

```
    printf("Enter a sentence: ");
```

```
    fgets(str, sizeof(str), stdin);
```

```
    printf("Number of words = %d", countWords(str));
```

```
    return 0;
```

```
}
```